

Recurrent Respiratory Papillomatosis: Juvenile Versus Adult Forms

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Recurrent respiratory papillomatosis (RRP) has been described to have a juvenile or aggressive form and an adult or less aggressive form. However, the aggressive form may occur in an adult and vice versa. Some authors have reported a quiescence in the juvenile form with the onset of puberty. In order to further characterize these two forms of RRP and to analyze the effects of puberty, we reviewed the records of 32 patients treated for RRP at our institution over a 10-year period. We found that the aggressive form typically occurs in the very youngest of patients (average of 2 years old as compared to an average of 17 years old in the less aggressive form). Although subglottic involvement universally occurred in our group with aggressive disease, approximately 40% developed subglottic disease very early as compared with 20% of patients with less aggressive disease. Additionally, our data do not support the theory of spontaneous regression with the onset of puberty. These and other findings will be discussed in detail. We also propose a new classification for RRP to eliminate confusion.

INTRODUCTION

Recurrent respiratory papillomatosis (RRP) is one of the most difficult and frustrating benign diseases that otolaryngologists treat. Papillomas can involve the entire aerodigestive tract but most often involve the glottis. They also constitute the most common benign neoplasm of the larynx. As the name implies, this disease can be aggressive and recurrent; it is found in all age groups and in both genders and can progress to malignancy. The etiology of this disease is the human papilloma virus (HPV). There are almost 70 different types of HPV which present with disease states ranging from cutaneous warts to cervical carcinoma. There has been an association between maternal cervical HPV infection and the incidence of RRP.¹ The same HPV types that have been found in

cervical condylomata have also been reported in laryngeal papillomatosis.²

Presenting symptoms of RRP most often include a change in voice, hoarseness, stridor, aphonia, and respiratory distress. Diagnostic work-up should include a thorough history and physical examination, laryngoscopy and/or bronchoscopy, and chest x-ray. The current treatment of choice is laryngoscopy and bronchoscopy with laser excision. This provides a precise method of excision combined with good hemostasis for this often friable lesion.

Complications of this disease and its treatment include stenosis, webbing, recurrence, vocal cord scarring, granulation tissue formation, tracheal perforation, tracheoesophageal fistula, respiratory arrest, bronchitis, pneumonia, pneumothorax, hemorrhage, and death.

Spontaneous resolution of disease with puberty has been reported, but remains controversial.^{3,4} Tracheotomy has also been discussed in detail by authors and has been found to be associated with spread to the trachea, bronchi, and lungs.^{5,6} Some authors believe that the patients requiring tracheotomy have a more aggressive disease form which would have led to distal spread regardless of whether the patient had a tracheotomy.⁴ Despite some difference of opinions on the cause and/or effect of tracheotomy, all authors agree that, if possible, a tracheotomy should be avoided. If a tracheotomy is necessary, decannulation should take place as early as possible.

As previously described, there are two different forms of this disease: a juvenile or aggressive form, and an adult or less aggressive form. Twenty years of age has been used as the age for adult onset, but the classification is not based purely on age.⁷ The aggressive form may occur in an adult and vice versa. If an adult acquires the aggressive form, he/she is said to have juvenile laryngeal papillomatosis. If a child develops the less aggressive form, he/she still has juvenile laryngeal papillomatosis. The classifications, as described, can be contradictory and confusing. We present the following study of 32 patients with RRP in order to better define the distinction between aggressive and nonaggressive disease. Additionally, we

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propose new terminology—aggressive and nonaggressive disease—to replace the prior terminology—juvenile and adult disease.

MATERIALS AND METHODS

We reviewed the records of all patients with the diagnosis of laryngeal papillomatosis at the Eye, Ear, Nose and Throat (EENT) Hospital in New Orleans, La., from 1980 to 1990. Forty-one patients were identified. All 41 patient charts were obtained and reviewed. Nine of the patients' diagnoses had been incorrectly coded; the other 32 patients were the basis for our review. All patients were assessed by microsuspension laryngoscopy and bronchoscopy for the presence or absence of glottic, subglottic, and tracheobronchial disease. Pulmonary disease was assessed via preoperative chest radiograph. Subglottic disease was defined as extension of disease distal to the free margin of the vocal cords but not beyond the lower border of the cricoid cartilage. Tracheal disease was defined as the extension of disease distal to the lower border of the cricoid. Bronchial disease was defined as below the carina.⁵ The charts were reviewed for patient demographics, age at diagnosis, number of procedures, length of treatment, location and number of sites of disease, outcome, complications, regression at puberty, and history of tracheotomy.

RESULTS

A total of 32 patients with a diagnosis of laryngeal papillomatosis were treated at the EENT hospital over a 10-year period. Eight of the patients did not have sufficient data to fully assess. Although some had a long history of disease, they had only one procedure performed at the EENT hospital with no follow-up. These 8 patients were not included in the study. RRP was defined as being aggressive in patients requiring greater than or equal to a total of 10 procedures and 3 or more procedures over a 1-year period or spread distal to the subglottic region. Patients without distal spread were not precluded from having "aggressive" disease. Nonaggressive disease was defined as being in patients requiring less than a total of 10 procedures, with no more than 3 procedures in a 1-year period, and absence of spread distal to the subglottis. Using these criteria, 14 patients had aggressive disease and 10 had nonaggressive disease. Using the classifications adult and juvenile, adults being 20 years or older and juveniles being under 20 years of age, 4 patients were adults and 20 were children. Of the patients with onset of disease at 20 years of age or older, all had nonaggressive disease. Of the patients with onset at under 20 years of age, 6 had nonaggressive disease and 14 had aggressive disease.

Demographics

Of the 24 patients included in the study, 13 were male and 11 were female, 13 were black and 11 were white. Three of the 4 adult patients were men, 3 were black and 1 was white. Of the 20 children, 10 were male and 10 were black. Six of the 14 patients with aggressive disease were female, 7 were black and 7 were white. The nonaggressive disease group consisted

of 5 females and 5 males, 6 black patients and 4 white patients.

Age at Onset

The age of onset of the patients ranged from 2 months to 39 years with an average of 8.7 years of age. Adult age of onset ranged from 31 to 39 years with an average of 36 years. The juvenile age of onset ranged from 2 months to 10 years with an average of 3.3 years. The average age at onset for patients with aggressive disease was 2.4 years with a range of 2 months to 10 years of age. Seventeen and one half years was the average age of onset for nonaggressive disease, with a range of 3 to 39 years.

Number of Procedures

The 20 patients with age of onset less than 20 years of age required 321 procedures with an average of 16 procedures per patient. The 4 patients 20 years of age or above required 18 procedures with an average 4.5 procedures per patient. The 14 patients with aggressive disease required 282 procedures with an average of 20 procedures per patient. The 10 patients with nonaggressive disease underwent 45 procedures, averaging 4.5 procedures per patient.

Early Subglottic Disease

The patients were assessed for the presence of subglottic spread during the first 6 months of their disease process. Over that time period, 1 of 4 adult patients was found to have subglottic disease and 7 of 20 juvenile patients had subglottic disease. Of patients with nonaggressive disease, 2 of 10 developed subglottic disease in the first 6 months of evaluation compared to 6 of 14 with aggressive disease.

Ultimate Subglottic Disease

Two of 4 adult patients developed subglottic disease during their course of disease. Fifteen of 20 children developed subglottic disease during their course of disease. Five of 10 patients with nonaggressive disease and 12 of 14 patients with aggressive disease developed subglottic disease.

Single Versus Multiple Site at Presentation

Upon presentation to the EENT hospital, 1 adult patient had single-site (glottic) disease, 2 adults had two sites of disease, and 1 adult had three sites of disease. Of the 20 children who were assessed upon presentation, 8 had single-site disease, 7 had two sites, 4 had three sites, and 1 had four sites. Of the patients with nonaggressive disease, 5 had one disease site, 4 had two disease sites, and 1 had three disease sites at presentation to our hospital. Of the patients with aggressive disease, 3 had single-site disease, 6 had two sites of disease, 4 had three disease sites, and 1 had four sites identified. In 1 of the patients with aggressive disease who had two sites of

TABLE I.
Sites of Involvement.

Sites	Juvenile (n = 20)	Adult (n = 4)	Nonaggressive (n = 10)	Aggressive (n = 14)
Supraglottic	17	3	6	14
Glottic	20	4	10	14
Subglottic	15	2	5	12
Tracheal	10	0	0	10
Bronchial	1	0	0	1
Pulmonary	1	0	0	1
Other	2	0	0	2

*Other sites included oral cavity and nasopharynx.

disease, oral cavity disease was present at one site. The patient with four sites of disease had two extralaryngeal sites, the nasopharynx and oral cavity. These were the only two patients in the study found to have disease superior to the larynx at any time during their course of disease (Table I).

Ultimate Number of Sites

During the patients' course of disease, one of four adults had one site of disease only, one had two total disease sites, and two had three disease sites, with an average of 2.25 sites of disease per adult. Two children had one site of disease, five had two sites, five had three sites, and eight had four sites, averaging three sites per child. Three patients with nonaggressive disease had one site of disease, four had two sites, and three had three sites, averaging two sites of disease per patient. Of the patients with aggressive disease, two had two sites of disease, four had three sites of disease, and eight had four sites of disease. There was an average of 3.4 disease sites per patient with aggressive disease.

Distal Spread

None of the patients with onset at 20 years of age or older had disease distal to the subglottis, whereas 11 of 20 patients with onset before the age of 20 had subglottic, tracheal, or pulmonary involvement. Eleven (79%) of 14 patients with aggressive disease had distal spread, whereas none of patients with nonaggressive disease had distal spread.

Outcome

Of the 4 adult patients, 1 had resolution of disease, 2 had minimal glottic disease, and 1 had glottic and subglottic disease at the time of last follow-up. Of the 10 patients with nonaggressive disease, 4 had resolution of disease, 5 had glottic disease, and 1 had glottic and subglottic disease at last follow-up. Of the 20 patients with onset of disease before the age of 20, 6 had resolution of disease and 14 had recurrence ranging from glottic disease to pulmonary disease. Three patients with aggressive disease had resolution of disease; the other 11 patients with aggressive disease had recurrence ranging from glottic to pulmonary

TABLE II.
Results.

	Juvenile	Adult	Aggressive	Nonaggressive
Gender	10 M, 10 F	3 M, 1 F	8 M, 6 F	5 M, 5 F
Onset age*	3.3 y	36 y	2.4 y	17.5 y
Procedure no.*	16	3.6	20	4.5
Early subglottic	35%	25%	43%	20%
Ultimate subglottic	75%	50%	86%	50%
Early site no.*	1.9	2.0	2.2	1.6
Ultimate site no.*	3.0	2.3	3.4	2.0
Distal spread	55%	0%	79%	0%
Resolution	30%	25%	21%	40%
Complications	75%	0%	93%	20%
Complications/patient*	2.1	0	2.9	0.2
Tracheotomy	60%	0%	79%	10%

*Averages.

disease. There were no deaths due to disease among the study patients.

Complications

Complications among the study patients, not including recurrence, consisted of anterior and posterior commissure webbing, glottic, subglottic, and tracheal stenosis, tracheal obstruction, stomal papillomas, vocal cord fusion, pneumomediastinum, and cardiac arrest. No adult patients in the study had complications. Five of 20 children had no complications. Fifteen children had anywhere from 1 to 7 complications with an average of 2.1 complications per child. Of the patients with nonaggressive disease, 2 of 10 had complications, both having just 1 complication, averaging 0.2 complications per patient. Patients with aggressive disease had a range of 0 to 7 complications with an average of 2.9 complications per patient. The 1 patient with aggressive disease who had no complications had pulmonary involvement.

Pubertal Regression

None of the nine patients who reached puberty had spontaneous regression of disease.

Tracheotomy

No adult patients required tracheotomy. Twelve of 20 children required tracheotomy. All were under 11 years old and all but 1 had aggressive disease. The 1 patient with nonaggressive disease who required a tracheotomy underwent an emergency tracheotomy because she was unable to be intubated at the time of surgery due to supraglottic and glottic disease. None of the patients had tracheal, bronchial, or pulmonary disease prior to tracheotomy. Ten patients had distal spread post-tracheotomy; 3 had resolution of the distal spread with the tracheotomy in place. Three of the 12 patients were decannulated; 1 patient had subsequent resolution of the distal spread postdecannulation.

Follow-up

The 24 patients were followed at the EENT hospital for 3 to 252 months with an average follow-up of 66 months.

DISCUSSION

Laryngeal papillomatosis remains an extremely difficult and frustrating disease to treat. There have been multiple medical and surgical treatment modalities with some success, more often for control rather than cure. The most common classifications have been juvenile and adult forms of the disease. Juvenile-onset disease is a more aggressive disease process; adult-onset disease is less aggressive. An adult can have aggressive disease, but the term juvenile form is used regardless of the age at onset. These classifications include a large gray area which can lead to confusion. Thirty-three percent of our juvenile patients who by definition of the classification should have had aggressive disease actually had nonaggressive disease. Our review was performed in order to look at the multiple characteristics involved in this disease and to find factors that may enable us to better classify this disease in the future (Table II).

We found a small male and black predominance in the adult population, otherwise there was little difference in sex and race between the juvenile patients, and the patients with aggressive or nonaggressive disease. Several authors have noted a male predominance of adults in their studies.^{4,8}

The average age of onset for all patients was 8.7 years. Our adult group was limited in size but had an average age of onset of 36 years. The juvenile group had an average age of onset of 3.3 years. Irwin, *et al.*¹⁴ found 83% of the patients they reviewed to have their onset of disease at or prior to the age of 4. Cohen, *et al.*⁹ noted 68% of their patients to present by age 4. Seventy percent of our patients presented at age 4 or younger. In the present study, the patients with nonaggressive disease presented, on average, at the age of 17.5 years; the patients with aggressive disease had an average age of onset of 2.4 years.

Adults required an average of 3.6 procedures per patient compared to 16 procedures per patient for children. Benjamin and Parsons reported an incidence of 13.7 procedures per adult and 30.7 procedures per child.³ Cohen, *et al.*⁹ reported 11.8 procedures per child and Morgan and Zitsch¹⁰ reported 12 procedures per child. In the present study, patients with aggressive disease required, on average, 20 procedures per patient, while those with nonaggressive disease required an average of 4.5 procedures per patient. Quiney, *et al.*¹¹ reported patients with less aggressive to require approximately 4 procedures and those with multiple confluent (more aggressive) disease to require 12.5 procedures.

Early subglottic involvement, *i.e.*, involvement

within 6 months of our initial evaluation, occurred in 25% of adults, 35% of children, 20% of those with nonaggressive disease, and 43% of those with aggressive disease. Early subglottic disease in aggressive patients has not been reported in the literature. Over the entire course of treatment, subglottic disease developed in 50% of the adult patients, 75% of the children, 50% of those with nonaggressive disease, and 86% of those with aggressive disease. This is consistent with Weiss and Kashima, who found subglottic disease present in 74% of their juvenile patients and in 58% of their adult patients.⁵

With regard to disease sites at our first evaluation, adults averaged 2 sites per patient and children averaged 1.9 sites per patient. Patients with nonaggressive disease averaged 1.6 sites of disease and those with aggressive disease averaged 2.2 sites of disease. The average total number of sites per adult was 2.3; for children, it was 3. For patients with nonaggressive disease, the average number of disease sites during their course of disease was 2 sites per patient; for those with aggressive disease, the average was 3.4 disease sites. Lindeberg, *et al.*⁷ believed that the fixed age of 20 as the borderline for adult disease was too rigid and proposed separating the groups into those with single and multiple lesions. As just described, we found little difference in the number of sites between adults and children both in early and late disease.

None of our adults had spread distal to the subglottic region, whereas 55% of our juvenile patients had distal spread. None of our nonaggressive disease patients had distal spread by definition, whereas 79% of patients with aggressive disease had spread distal to the subglottis. Weiss and Kashima reported a similar incidence of distal spread between their adult and juvenile patients, 25% and 26%, respectively.⁵

In the present study, 25% of the adult patients and 30% of the juvenile patients had resolution of disease. Forty percent of patients with nonaggressive disease and 21% of patients with aggressive disease had resolution. Cohen, *et al.* reported a resolution rate of approximately 30% for their patients.⁹

No adults had complications. Seventy-five percent of the children had complications ranging from webbing to cardiac arrest, with an average of 2.1 complications per patient. Patients with nonaggressive disease averaged 0.2 complications per patient, whereas patients with aggressive disease averaged 2.9 complications per patient.

No adults had disease severe enough to require tracheotomy, but 60% of the children underwent tracheotomy for impending or actual respiratory distress. The incidence of tracheotomy has been described by many authors with ranges of 1.8% to 64%.^{4,5,9,12-14} In the present study, 1 (10%) of the patients with nonaggressive disease had a tracheotomy, while 11 (79%) of the 14 patients with aggressive

disease required tracheotomy. Tracheotomy was performed within the first year of disease for 10 of the 12 patients requiring tracheotomy. The patient with non-aggressive disease who underwent tracheotomy was decannulated after 14 months and the disease then resolved. None of the patients had distal spread (below the subglottis) prior to tracheotomy, but 10 of the 11 patients with aggressive disease developed distal disease post-tracheotomy. Two of the patients with aggressive disease were decannulated; 1 was decannulated 3 months post-tracheotomy and the other was decannulated 7 years post-tracheotomy. The patient who was decannulated after 7 years had resolution of disease. The patient who was decannulated after 3 months had required a prior tracheotomy and did not have resolution of disease.

As previously described, we found no signs of regression in the nine patients in this review who reached puberty.

We found that the patients with more aggressive disease were patients usually requiring more procedures more often, often required tracheotomy, had more complications associated with the disease, and often had spread distal to the subglottis. These patients were more likely to have uncontrollable disease and less likely to have their disease resolve. Characteristics which were consistent for patients with more aggressive disease included a total of 10 or more procedures with 3 or more procedures within a 1-year period and a high incidence of spread distal to the subglottis. In patients with less aggressive disease, none had distal spread, all had less than 3 procedures

over a 1-year period, and all required less than a total of 10 procedures at our institution. These patients were unlikely to require a tracheotomy, had less procedures performed, incurred less complications, and were more likely to have resolution or control of disease.

This is a retrospective review and the numbers are small, but our proposal to reclassify laryngeal papillomatosis as aggressive and nonaggressive is more reflective of the actual disease process than the original classifications of adult and juvenile. Further studies should be performed with larger patient populations with this disease in order to validate these findings.

CONCLUSION

In summary, we have proposed new classifications for laryngeal papillomatosis—aggressive and nonaggressive—to replace the past classifications of juvenile and adult. The characteristics of aggressive disease include 10 or more total procedures with 3 or more procedures within a 1-year period and/or spread of disease distal to the subglottis. In contrast, the characteristics of nonaggressive disease include less than 10 total procedures, less than 3 procedures within a 1-year period, and the absence of distal spread. We also found early subglottic disease to be more prevalent in the patients with aggressive disease, and ultimately subglottic disease was much more common in aggressive than in nonaggressive disease. We found no regression associated with puberty.

BIBLIOGRAPHY

1. Bennet, R.S. and Powell, K.R.: Human Papillomaviruses: Associations Between Laryngeal Papillomas and Genital Warts. *Pediatr Infect Dis J*, 6:22–32, 1987.
2. Abramson, A.L., Steinberg, B.M. and Winkler, B.: Laryngeal Papillomatosis: Clinical, Histopathologic and Molecular Studies. *aryngoscope*, 97:678–685, 1987.
3. Benjamin, B. and Parsons, D.S.: Recurrent Respiratory Papillomatosis: A 10-Year Study. *J Laryngol Otol*, 102:1022–1028, 1988.
4. Majoros, M., Parkhill, E.M. and Devine, K.D.: Papilloma of the Larynx in Children. *Am J Surg*, 108:470–475, 1964.
5. Weiss, M.D. and Kashima, H.K.: Tracheal Involvement in Laryngeal Papillomatosis. *LARYNGOSCOPE*, 93:45–48, 1983.
6. Cole, R.R., Myer, C.M. and Cotton, R.T.: Tracheotomy in Children With Recurrent Respiratory Papillomatosis. *Head Neck*, 11:226–230, 1989.
7. Lindeberg, H., Oster, S., Oxlund, I., et al.: Laryngeal Papillomas: Classification and Course. *Clin Otolaryngol*, 11:423–429, 1986.
8. Capper, J.W., Bailey, C.M. and Michaels, L.: Squamous Papilloma of the Larynx in Adults. *Clin Otolaryngol*, 8:109–119, 1983.
9. Cohen, S.R., Geller, K.A., Seltzer, S., et al.: Papilloma of the Larynx and Tracheobronchial Tree in Children: A Retrospective Study. *Ann Otol Rhinol Laryngol*, 89:497–503, 1980.
10. Morgan, A.H. and Zitsch, R.P.: Recurrent Respiratory Papillomatosis in Children: A Retrospective Study of Management of Complications. *Ear Nose Throat J*, 65:403–410, 1986.
11. Quiney, R.E., Hall, D. and Croft, C.B.: Laryngeal Papillomatosis: Analysis of 113 Patients. *Clin Otolaryngol*, 14:217–225, 1989.
12. Dedo, H.H. and Jackler, R.K.: Laryngeal Papilloma: Results of Treatment With CO₂ Laser and Podophyllum. *Ann Otol Rhinol Laryngol*, 91:425–430, 1982.
13. Strong, M.S., Vaughan, C.W., Cooperbrand, S.R., et al.: Recurrent Respiratory Papillomatosis. *Ann Otol Rhinol Laryngol*, 85:508–516, 1976.
14. Irwin, B.C., Hendricks, W.A., Pincott, J.R., et al.: Juvenile Laryngeal Papillomatosis. *J Laryngol Otol*, 100:435–445, 1986.